

What is an API?

API stands for "Application Programming Interface." In simple terms, it's a set of rules and protocols that allow how different software applications can communicate and interact with each other. APIs define the methods and data formats that applications can use to request and exchange information. To retrieve data from a web server, a client application initiates a request, and the server responds with the requested data.

APIs act as bridges that enable the smooth exchange of data and functionality, enhancing interoperability across various applications.

Making API Requests in Python

In order to work with API some tools are required such as requests so we need to first install them in our system.

Command to install 'requests':

➤ **pip install requests**

Once we have installed it, we need to import it in our code to use it.

Command to import 'requests':

Making API calls

➤ `var = requests.get(url)`

Explanation:

- ✓ Builds a URL with parameters including API key.
- ✓ Sends GET request to NewsAPI.
- ✓ Prints status code to confirm request success (200 means OK).

JSON Data

JSON is the common format APIs use to send data. Think of JSON as the language APIs speak to share information like news headlines, images, and descriptions.

To handle this data, we use Python and the requests library to fetch news from NewsAPI. Then, Python helps organize and display the top articles clearly- like a helpful librarian sorting through lots of information for us. This shows how JSON and Python work together to make API data easy to use.

Understanding API Status Codes

Status codes tell us how the server handled our request:

200 OK: Request successful, data returned.

201 Created: New resource created.

204 No Content: Success but no data returned.

400 Bad Request: Invalid request.

401 Unauthorized: Missing or invalid API key.

500 Internal Server Error: Server encountered an error.

Example

```
import requests
```

```
def weather_details(city, api_key):
```

```
    url = "https://api.openweathermap.org/data/2.5/weather?"
```

```
    final_url = f"{url}q={city }&appid={api_key}&units=metric"
```

```
    try:
```

```
        response = requests.get(final_url)
```

```
        response.raise_for_status()
```

```
        data = response.json()
```

```
        temperature = data["main"]["temp"]
```

```
        humidity = data["main"]["humidity"]
```

```
        feels_like = data["main"]["feels_like"]
```

```
        print(f"Weather in {city}:")
```

```
        print(f"Temperature: {temperature}°C (Feels like: {feels_like}°C)")
```

```
        print(f"Humidity: {humidity}%")
```

```
        print(f"Description: {data['weather'][0]['description'].capitalize()}")
```

```
    except requests.exceptions.RequestException as e:
```

```
        print(f"Error fetching weather data: {e}")
```

```
api_key = "b8aeb299f93d443bd5cbeecdb72fc733"
```

```
city = input("Enter city name: ")
```

```
weather_details(city, api_key)
```

PYTHON WHATSAPP MESSAGING

Install whatsapp Package using command

```
pip3 install pywhatkit
```

import it in your program

```
import pywhatkit
```

command to send message

```
pywhatkit.sendwhatmsg('mobile no', 'message', time_hour, time_min)
```

Random Example

ROCK PAPER SCISSOR GAME

```
import random
```

```
l = ["rock", "paper", "scissor"]
```

```
while True:
```

```
    c = int(input("Game Start.....
```

```
1 YES
```

```
2 NO | Exit"))
```

```
    if c == 1:
```

```
        uscore = 0
```

```
        cscore = 0
```

```
        for a in range(1, 6):
```

```
            uch = 0
```

```
            userinput = int(input("
```

```
1 ROCK
```

```
2 PAPER
```

```
3 SCISSOR"))
```

```
if userinput == 1:
    uch = "rock"
elif userinput == 2:
    uch = "paper"
elif userinput == 3:
    uch = "scissor"
else:
    print("invalid choice")
cch = random.choice(l)
print("Your Choice ", uch, "Computer Choice", cch)
if uch == cch:
    print("Game Draw")
    uscore += 1
    cscore += 1
elif ((uch == "rock" and cch == "scissor") or
      (uch == "paper" and cch == "rock") or
      (uch == "scissor" and cch == "paper")):
    uscore += 2
    print("You Win")
else:
```



```
        cscore += 2

        print("Computer Win")

    if uscore == cscore:

        print("Game Drawn")

    elif uscore > cscore:

        print("You Won")

    elif cscore > uscore:

        print("Computer Won")

else:

    break
```